

# Final Project Presentation

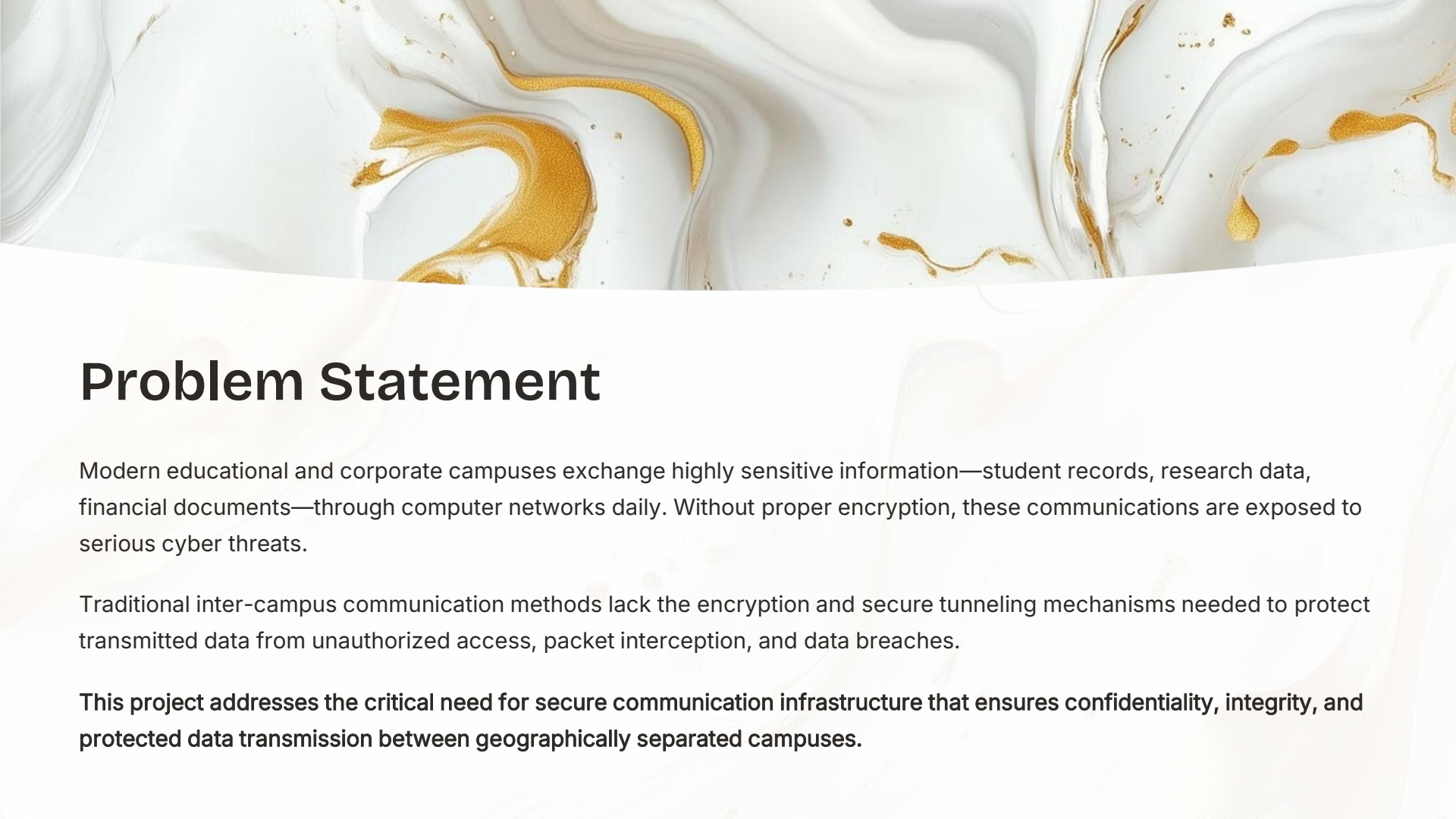
## Establishing Encrypted Communication Between Two Campuses Using Industry- Standard Encryption Policies

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Internship Domain: Cybersecurity & Ethical Hacking

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# Problem Statement

Modern educational and corporate campuses exchange highly sensitive information—student records, research data, financial documents—through computer networks daily. Without proper encryption, these communications are exposed to serious cyber threats.

Traditional inter-campus communication methods lack the encryption and secure tunneling mechanisms needed to protect transmitted data from unauthorized access, packet interception, and data breaches.

**This project addresses the critical need for secure communication infrastructure that ensures confidentiality, integrity, and protected data transmission between geographically separated campuses.**

# Project Objective



## Establish Secure Communication

Connect two campuses with encrypted channels



## Implement Site-to-Site IPSec VPN

Deploy industry-standard tunneling protocol



## Configure AES and SHA

Apply encryption standards for security



## Secure WAN Connectivity

Enable protected wide area network links



## Verify Encrypted Transmission

Test and validate packet encryption

# Proposed Solution

01

## Secure Network Infrastructure Design

Engineered encrypted communication between geographically separated campuses using routers, switches, PCs, and WAN serial connections simulated in Cisco Packet Tracer

02

## Site-to-Site IPSec VPN Implementation

Established encrypted tunnel between campus routers to secure all inter-campus communication channels

03

## Industry-Standard Encryption

Configured AES encryption and SHA hashing protocols to ensure data confidentiality and message integrity

04

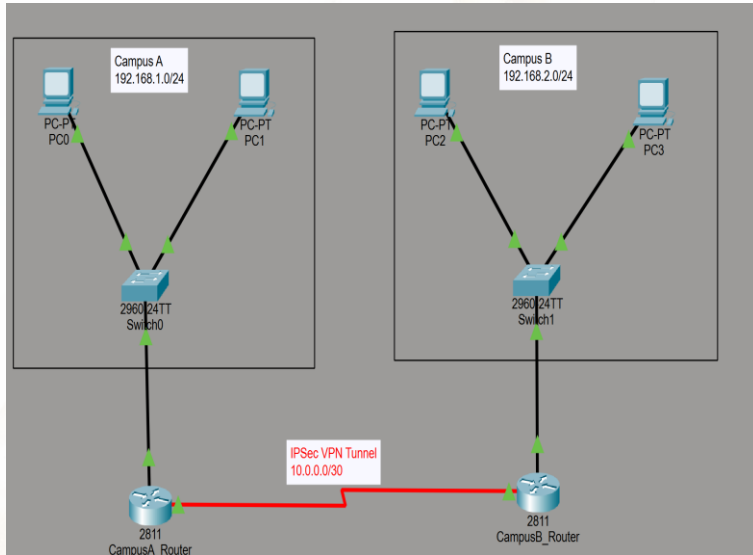
## Static Routing Configuration

Enabled secure path determination for communication between campus networks through the VPN tunnel



# Implemented Network Topology

The implemented topology simulates secure communication between two geographically separated campuses using IP addressing schemes and virtual routing infrastructure.



## Campus A Network

192.168.1.0/24  
subnet with local  
devices and router  
gateway

## Campus B Network

192.168.2.0/24  
subnet with local  
devices and router  
gateway

## WAN Connection

Serial link  
connecting routers  
through secure VPN  
tunnel



# Implementation and Configuration



## Router Configuration

Initialized interfaces and IP addressing on both campus routers



## WAN Serial Configuration

Established physical layer connectivity between router interfaces



## Static Routing

Configured routing tables to direct traffic through designated paths



## IPSec VPN Configuration

Implemented encryption parameters and tunnel specifications



## Crypto Map Implementation

Applied security policies to router interfaces for packet protection

**Configuration Tools:** Cisco Packet Tracer for network simulation, CLI commands for router configuration, and security protocols for encryption implementation

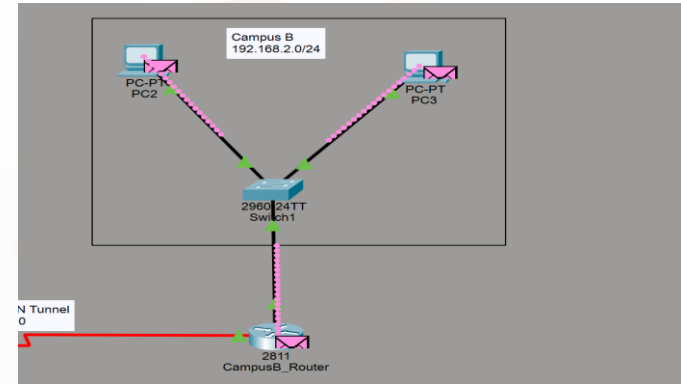
# Connectivity Testing and Live Demonstration

## Test Results

- Successful communication established between Campus A and Campus B networks
- IPSec VPN tunnel successfully carried encrypted traffic across WAN connection
- Inter-campus connectivity verified using ICMP ping testing procedures
- VPN tunnel activity validated using security association verification commands
- Simulation mode used to analyze real-time packet flow through secure tunnel

```
Request timed out.  
Request timed out.  
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126  
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126  
  
Ping statistics for 192.168.2.10:  
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

## Successful Ping Test



## Packet Simulation Analysis

Visualized packet movement showing encrypted data traveling between campuses through VPN path

**Result:** Secure tunnel routing validated

# VPN Verification and Security Validation

Security validation commands confirmed successful IPsec VPN tunnel establishment and encrypted packet transmission.

## show crypto isakmp sa

Displays Internet Security Association and Key Management Protocol security associations

```
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.0.2      10.0.0.1      QM_IDLE        1096      0 ACTIVE

IPv6 Crypto ISAKMP SA
```

**QM\_IDLE state confirms successful IPsec VPN tunnel establishment.**

## show crypto ipsec sa

```
Router#show crypto ipsec sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.0.2      10.0.0.1      QM_IDLE        1096      0 ACTIVE

IPv6 Crypto ISAKMP SA

Router#show crypto ipsec sa
interface: Serial0/3/0
Crypto map tag: VPNMAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
#pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/3/0
current outbound spi: 0x0(0)

inbound esp sas:
```

**Encrypted packet transmission verified successfully.**



# Results and Achievements

## Secure Encrypted Communication

Successfully established encrypted communication between Campus A and Campus B networks

## IPSec VPN Tunnel

Site-to-Site IPSec VPN tunnel successfully configured and verified through testing

## AES and SHA Encryption

Implemented AES encryption and SHA hashing mechanisms for secure data transmission

## Verified Connectivity

Successful inter-campus communication verified using comprehensive connectivity and VPN testing

## Packet Validation

Secure packet transmission through VPN tunnel validated using IPSec security association commands

## Simulation Demonstration

Simulation mode successfully demonstrated packet flow through encrypted tunnel

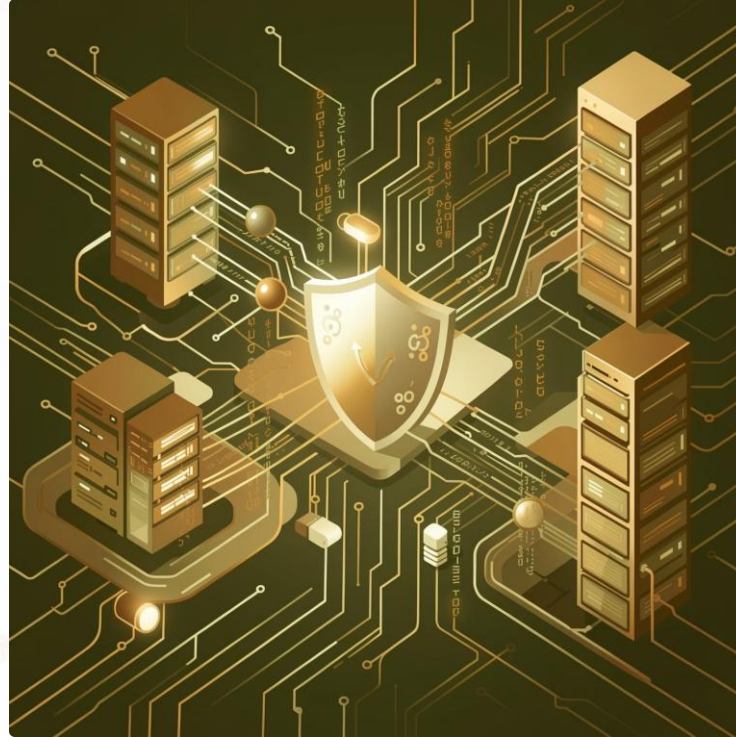
✔ **Project Success:** The project successfully achieved secure communication between two remote campuses using IPSec VPN with industry-standard encryption policies. Connectivity, encryption, and VPN tunnel establishment were successfully tested and validated.

# Conclusion

This project successfully demonstrated the implementation of encrypted communication between two geographically separated campuses using IPSec VPN with industry-standard encryption policies.

The Site-to-Site VPN configuration established a secure tunnel using AES encryption and SHA hashing, ensuring data confidentiality and integrity during transmission across WAN connections.

Testing confirmed successful packet encapsulation, encrypted transmission, and secure inter-campus connectivity, validating the implementation approach and security measures.



## Key Takeaways

- ▼ Industry-standard IPSec protocols provide reliable encryption for campus networks
- ▼ Proper configuration of routers and security parameters is essential for tunnel establishment
- ▼ Testing and verification commands confirm secure operation and encrypted packet transmission